

Better Gate Model — Alt Training Report

Generated 2026-08-29 21:10

Dataset /home/malnitro5/Documents/Projects/tyndall/Ikmal/Gate Model/Better Gate Model/leaf_nonleaf_data

Overall accuracy (clean test)

95.78%

3310 / 3456

Camera-stress accuracy

96.04%

3319 / 3456

Best EMA validation

96.36%

Target

92%

PASS

Per-class accuracy (recall)

leaf

clean

92.9%

stress

95.5%

P=0.985

F1=0.957

n=1728

non-leaf

clean

98.6%

stress

96.6%

P=0.933

F1=0.959

n=1728

FN_leaf (true leaf → non-leaf)

122

FP_leaf (true non-leaf → leaf)

24

Student TinyLeafGate

27,538 params

96×96

Teacher MobileNetV3-Small

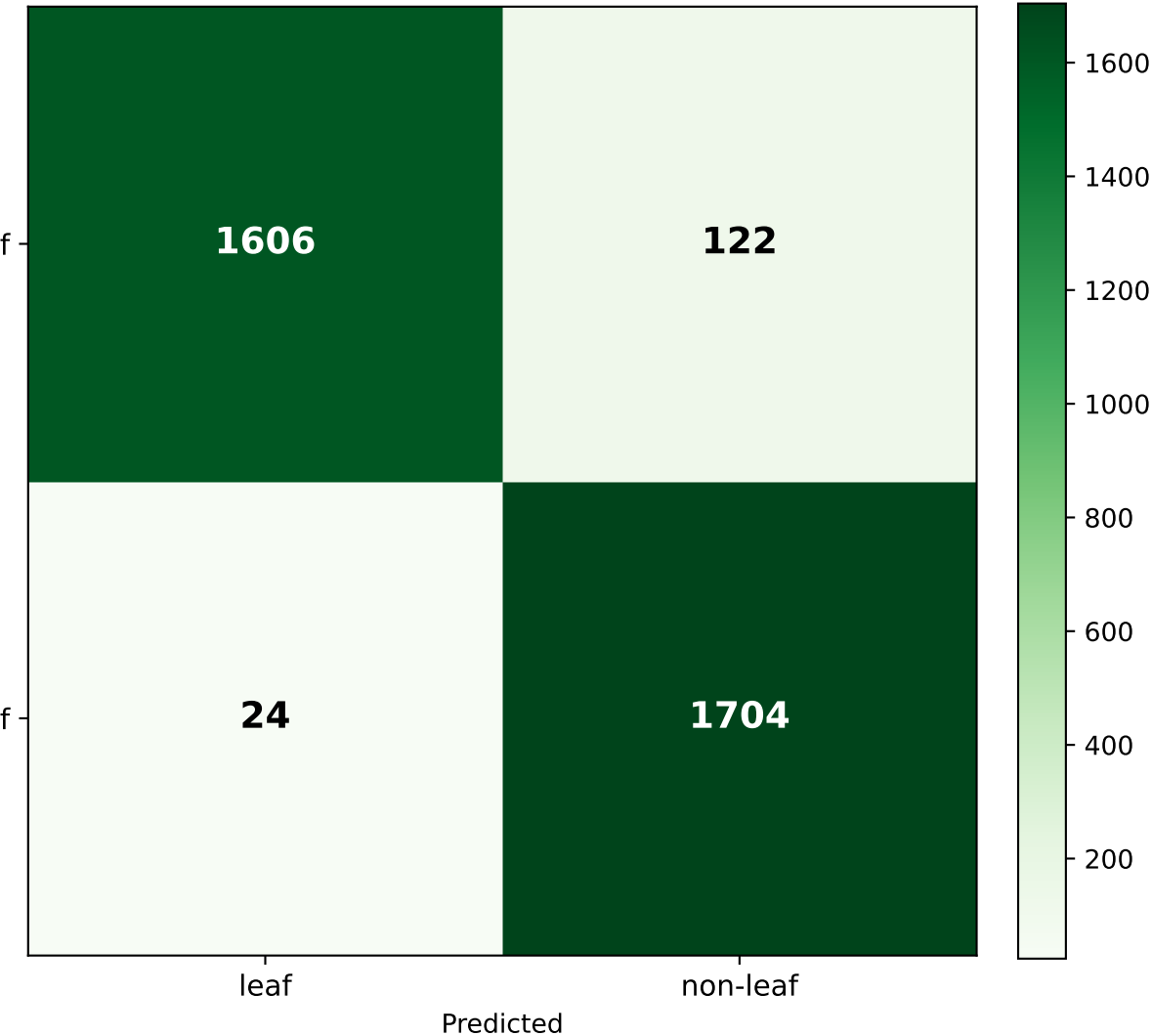
KD α=0.7

T=4.0

MixUp=0.2

EMA=0.999

Confusion matrix — clean PyTorch FP32 test



Dataset splits (balanced leaf / non-leaf)

Split	leaf	non-leaf	total
train	8,067	8,067	16,134
val	1,729	1,729	3,458
test	1,728	1,728	3,456

Pools were collected from PlantVillage first (leaf vs non-leaf),
balanced to the same count, then split into train/val/test.

Confusion matrix rows are true labels; columns are predictions.
Diagonal cells are correct counts.